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Conf-ReAct User-Controlled Tradeoffs in LLM Agent Reasoning via Conformal Prediction

 (/pdf?id=YMjQZVZmIV)

*Saptarshi Mondal (/profile?id=~Saptarshi_Mondal1),
Tiku Mohammd (/profile?id=~Tiku_Mohammd1),
Sanga Chaki (/profile?id=~Sanga_Chaki1),
Sourangshu Bhattacharya (/profile?id=~Sourangshu_Bhattacharya1) *

 21 Jul 2026 (modified: 25 Sept 2026)  Submitted to AAAI 2027

 Conference, Area Chairs, Senior Program Committee, Program Committee, Authors

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TL;DR: Conf-ReAct User-Controlled Tradeoffs in LLM Agent Reasoning via Conformal Prediction

Abstract:

Large language models (LLMs) are increasingly used as decision-making and planning agents in interactive text-based reasoning and game-playing environments, where they must repeatedly reason about an evolving state and select actions under uncertainty. Existing approaches largely rely on the LLM's own confidence signals or iterative self-critique (e.g., ReAct, Reflexion) to guide this process, offering little formal guarantee that the agent's beliefs about the environment or its decisions are trustworthy. We introduce Conf-ReAct, a conformal-prediction-based framework that equips LLM-driven agents with a statistically calibrated confidence set over plausible next actions at each state. We introduce a learned nonconformity score function that maps each (state, action) pair to a calibrated measure of fit, trained via binary cross-entropy against back-labeled trajectory outcomes. Using this score, we construct state-conditional conformal prediction sets over the LLM's candidate actions that adapt the pruning threshold to the local difficulty of the current state, retaining only a calibrated subset of actions for exploration via a lightweight breadth-first search. We evaluate Conf-ReAct on three interactive reasoning environments — WebShop, Game of 24, and ScienceWorld — using three open-source language models ranging from 3B to 12B parameters, comparing against ReAct, Reflexion, ReflAct, and Rollback. At a 90% target coverage level, Conf-ReAct improves task-success rate over the strongest baseline by up to 25% on WebShop, 35% on Game of 24, and 7% on ScienceWorld, while exposing a predictable, single-parameter (α) trade-off between success rate and inference cost.

Primary Topic: MAS: LLM-based Agents & Agentic Systems

Secondary Topics: ML: Bayesian Learning & Uncertainty Quantification

Country of Institutions:  India

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Paper Decision

Decision by Program Chairs 25 Sept 2026, 00:18 (modified: 25 Sept 2026, 04:46)

Program Chairs, Authors, Program Committee Submitted Revisions (/revisions?id=DCyDA6ketC)

Decision: Reject

Comment:

This paper did not advance to Phase 2 review for the AAAI-27 conference. The paper was reviewed by at least two human reviewers and at least one senior member of the program committee in Phase 1. In addition, papers with at least one positive review were examined by one program chair or associate program chair to ensure the rejection was justified. **This decision is final.** The provided reviews contain more detail, which we hope will be beneficial to the progress of this research.

Interesting conformal action-selection idea, but the claimed guarantees and cost comparisons are not fully convincing

Official Review by Program Committee 45Cd 01 Sept 2026, 16:47 (modified: 25 Sept 2026, 03:18)

Program Chairs, Area Chairs, Senior Program Committee, Program Committee 45Cd, Authors, Program Committee Submitted

Revisions (/revisions?id=13aW77O68y)

Paper Summary:

The paper proposes Conf-ReAct, a conformal-prediction-based extension of ReAct for LLM agents. Instead of committing to one action, the method generates multiple candidate actions, scores them with a learned nonconformity function, constructs a state-dependent conformal action set, and explores the retained actions using breadth-first search.

Strengths:

The paper addresses an important problem in LLM agents: balancing exploration, reliability, and inference cost. Using conformal prediction to control the number of candidate actions is intuitive and provides a useful user-facing parameter. The evaluation spans three different environments and three model sizes, and the reported success-rate improvements are substantial in several settings; for example, at 90% target coverage the paper reports gains of up to 25% on WebShop and 35% on Game of 24 over the strongest baseline. The state-dependent action-set construction is also more appealing than applying one global confidence threshold.

Weaknesses:

My main concern is that the paper's strongest motivation is statistical calibration/coverage guarantees, yet the required exchangeability assumption does not hold naturally in the proposed sequential setting. The calibration cache is generated offline using an auxiliary LLM, while test states are induced online by Conf-ReAct's own policy. The authors explicitly acknowledge that these distributions need not be exchangeable and therefore describe the measured coverage as only approximate rather than formally guaranteed. This substantially weakens the central theoretical claim.

The efficiency comparison is also incomplete. Conf-ReAct often uses substantially more LLM calls at high coverage—for example, 30.80 calls on WebShop with Qwen-2.5-3B at 90% coverage—and additionally explores multiple environment actions through BFS. Reporting mainly LLM calls per successful episode, rather than total cost over

all episodes, tokens, environment interactions, and wall-clock cost, makes it difficult to assess the actual success-cost trade-off fairly. Finally, performance and empirical coverage are noticeably weaker on Game of 24, suggesting sensitivity to the quality of the learned scoring function and state embeddings.

Justification of Recommendation:

The core idea is interesting, practically relevant, and experimentally promising. However, there is a significant gap between the paper's framing around conformal guarantees and what can actually be guaranteed under the policy-induced distribution shift. I would also like to see a more comprehensive cost evaluation, stronger ablations against simpler score-based/top-(k) pruning methods, and clearer analysis of the auxiliary-LLM/calibration distribution mismatch. These issues prevent me from being confident that the current contribution is sufficiently established.

Specific Points of Feedback for Rebuttal:

See weakness.

Rating: 5: Marginally below acceptance threshold

Confidence: 4: The reviewer is confident but not absolutely certain that the evaluation is correct

Review of Conf-ReAct

Official Review by Program Committee CWZr  26 Aug 2026, 08:59 (modified: 25 Sept 2026, 03:18)

 Program Chairs, Area Chairs, Senior Program Committee, Program Committee CWZr, Authors, Program Committee Submitted

 Revisions (/revisions?id=nkGOQCT7WA)

Paper Summary:

This paper proposes Conf-ReAct, a conformal-prediction-based framework for controlling the success-cost trade-off in LLM-agent planning. It trains a state-action scoring function from offline trajectories, applies localized conditional calibration to retain a subset of LLM-generated actions, and explores the resulting branches using breadth-first search. Experiments on WebShop, Game of 24, and ScienceWorld with three language models report improved task-success rates under different target-coverage settings.

Strengths:

The paper addresses the practically important trade-off between committing prematurely to a single action and exhaustively exploring multiple candidates. The proposed framework is conceptually clear, and the parameter α provides an intuitive way to adjust the breadth of exploration. The evaluation covers three different interactive environments and model sizes from 3B to 12B, with substantial success-rate improvements in several settings.

Weaknesses:

1. Conf-ReAct is essentially an incremental extension of existing conformal action pruning, replacing the global threshold used by Prune 'n Predict with localized conditional calibration and coupling the retained action set with conventional breadth-first search. The contribution therefore appears to be an integration of established conformal-prediction and search components rather than a substantially new agent-planning framework.
2. The paper describes the retained action set as being guaranteed to contain the optimal next action at the user-specified coverage level, but it later acknowledges that the offline auxiliary-LLM cache and the online policy-induced states need not be exchangeable; consequently, the reported coverage is only approximate. Moreover, a positive training label only means that one sampled trajectory containing the action eventually succeeded, rather than that the action was optimal, while the recall of the initial LLM-generated shortlist is not reported. These issues leave both the guarantee and its underlying coverage target insufficiently established.
3. Conf-ReAct receives an offline state-action cache constructed from auxiliary-LLM trajectories and success/failure supervision, and then explores multiple retained branches through environment execution and reset. The baselines do not appear to receive equivalent offline supervision, candidate expansions, or environment-interaction budgets; therefore, the reported success gains may partly reflect additional information and search access rather than a more effective action-selection rule.
4. The reported success improvements partly come from shifting the planning process from single-trajectory action selection to broader multi-branch execution. On WebShop with Qwen-2.5-3B, for example, Conf-ReAct at 90% target coverage reports 30.80 average LLM calls conditional on successful episodes, whereas Rollback and ReflAct report only 10.04 and 6.50 calls, respectively; the additional environment actions, resets, and state expansions introduced by BFS are not reported. Because Table 1 excludes failed episodes, this

conditional average cannot represent expected end-to-end cost, and the paper also omits token consumption, offline cache construction, and wall-clock latency. The current evidence therefore does not establish superior end-to-end cost efficiency, but instead suggests that the success gains are obtained by shifting part of the cost to broader controlled search. I do not dispute the reported success improvements, but I believe this trade-off substantially weakens their practical significance.

5. The current experiments vary only the target-coverage parameter and do not separately evaluate the learned scorer, localized calibration, candidate-set construction, and breadth-first search under matched expansion budgets. They therefore show that the complete system can improve success rates, but not which components improve the success-cost trade-off, whether conditional calibration is necessary, or whether similar performance could be obtained through simpler score-based search.

Justification of Recommendation:

The rebuttal should clarify what Conf-ReAct adds beyond Prune 'n Predict, what coverage statement remains valid under the online policy-induced state distribution and trajectory-based labels, and whether the reported advantage persists under matched offline supervision and end-to-end resource budgets that include failed episodes. Comparisons with global conformal pruning and simpler score-based search, together with component ablations under matched expansion budgets, would be particularly important.

Specific Points of Feedback for Rebuttal:

I have already detailed the specific issues that could affect my assessment in the Paper Weaknesses and Justification of Recommendation sections.

Rating: 4: Ok but not good enough - rejection

Confidence: 4: The reviewer is confident but not absolutely certain that the evaluation is correct

AI Review

AI Review by AI Reviewer (🤖) 📅 27 Jul 2026, 17:29 (modified: 25 Sept 2026, 03:18)

👁 Program Chairs, AI Reviewer, Area Chairs, Senior Program Committee, Authors, Program Committee
Submitted

📄 Revisions (/revisions?id=hmsTjiG4eV)

Review:

Title: Conf-ReAct: User-Controlled Tradeoffs in LLM Agent Reasoning via Conformal Prediction

Synopsis of the paper

The paper proposes Conf-ReAct, a framework that uses conformal prediction to retain state-dependent sets of candidate actions for an LLM agent. It learns action scores from offline trajectories, calibrates thresholds using nearby cached states, and explores retained branches with breadth-first search. Experiments across web navigation, arithmetic reasoning, and text-based science environments examine whether the approach improves task completion while exposing a user-controlled success-cost trade-off.

Summary of Review

The paper addresses the consequential problem of balancing search breadth and inference cost in sequential LLM agents, with localized calibration constituting its main distinction from prior global conformal pruning. The reported system-level success gains are substantial on WebShop and Game of 24, and task success and call counts respond predictably to the coverage parameter. However, the paper's own exchangeability caveat contradicts its deployed coverage guarantee, while the specified labels, calibration scores, and test-dependent localization do not establish coverage of a defined success-preserving action. The evaluation also cannot separate calibration benefits from broader BFS exploration, and its cost metric excludes failures and environment interactions.

Strengths

- The success-rate gains on WebShop and Game of 24 are large relative to differences among the listed baselines. At the 90% target in Table 1, gains over the strongest baseline range from 21 to 35 percentage points across backbones, indicating a substantial system-level effect rather than a narrow metric difference.
- The evaluation covers distinct decision regimes and multiple model scales. Section 4.1 includes web navigation, exact symbolic composition, and long-horizon embodied reasoning with three backbones, providing evidence that the observed behavior is not confined to one environment or model size.

- The control parameter behaves predictably on the principal task-level outcomes. Table 1 shows that decreasing α consistently increases success and LLM calls across all three environments and backbones, supporting the practical premise of an interpretable exploration-control interface.

Weaknesses

- The paper’s deployed coverage guarantee is contradicted by its own stated assumptions and measurements. Section 3.2 says $\hat{C}(s)$ is “guaranteed” at level $1 - \alpha$ to contain the optimal next action, but Section 3.3 says offline $D_{\text{cal}}(s_t)$ and online policy-induced s_t “need not be exchangeable” and that coverage is “approximate, rather than formally guaranteed”; Table 2’s 90%-target Game of 24 bins are 55–72% in several rows. This leaves the central guarantee unsupported in the evaluated regime.
- The calibration target is not defined in a way that yields action coverage. Section 3.3 labels an action positive when an auxiliary-LLM continuation succeeds, although a failed continuation does not establish that the first action cannot lead to success; it then defines D_{cache} over positive and negative triples while $V(s, a) = 1 - g_\theta(s, a)$ ignores y , and Equation (2) does not state that only positive-action scores are calibrated. Thus neither “optimal action” nor the score distribution whose quantile should be covered is identified.
- The cited conditional-calibration result does not, as presented, justify the test-dependent localization. Section 3.3 first selects the k nearest cached states to the current test state and then applies Equation (2) within that subset; selection based on the test state does not itself make the subset exchangeable with the test point, and no validity result is given for this added procedure. This issue is independent of the acknowledged auxiliary-policy/test-policy shift and directly affects the claimed state-conditional validity.
- Table 1 cannot attribute the gains to conditional conformal calibration because Conf-ReAct changes both action selection and search: Section 3.2 uses multi-branch FIFO BFS, whereas the listed baselines use single-branch action, reflection, or rollback procedures. No matched-budget top- k /scorer-guided BFS, execute-all-shortlist, or global-conformal ablation is reported; the cited global-conformal predecessor is not compared (Vishwakarma et al., 2025), and LATS is absent from the paper’s related work and tables (Zhou et al., 2024). Branching and learned scoring therefore remain confounded with localized calibration.
- The cost claim rests on a selected-sample metric. Table 1 defines LLM calls as average calls “per winning episode,” omitting failures that may exhaust the budget; the method also repeatedly resets states and executes retained actions, but environment actions, replay costs, and generated tokens are unreported. The claimed success–cost trade-off therefore cannot be interpreted as expected cost per task or compared under a common total budget.
- The ScienceWorld gains lack enough statistical information to assess stability. Table 1’s 90% setting reaches only 10%, 32%, and 44% absolute success and improves over the strongest baseline by 3, 5, and 7 percentage points, respectively, while evaluation-set sizes, repeated runs, and uncertainty intervals are not reported. These margins require uncertainty estimates to distinguish systematic gains from evaluation variation.

Suggestions for Improvement

- A trajectory-level reliability analysis would better connect per-state action sets to sequential task completion. Measuring how often an entire success-preserving path survives pruning as a function of horizon would expose error accumulation that per-state coverage cannot characterize.
- Section 3 would benefit from one offline/online algorithm box defining candidate generation, score-training and calibration splits, threshold computation, state restoration, queue expansion, and budget accounting. This would let readers distinguish the scorer, calibration rule, and search policy without reconstructing their interaction across Sections 3.2–3.3.
- An error decomposition separating shortlist omission, scorer-ranking error, pruning error, and search-budget exhaustion would clarify where the remaining failures arise and whether the dominant limitation differs across WebShop, Game of 24, and ScienceWorld.

Additional Details and Minor Issues

- Table 1’s parenthetical gains are percentage-point differences, not relative percentages; several half-point differences are truncated.
- Equation (3) can return an empty set despite Section 3.2 assuming $|\hat{C}(s)| \geq 1$; the fallback policy is unspecified.
- Section 4.3 says Figure 2 covers all three environments, but the figure contains only WebShop and Game of 24.

- Section 4.3 claims binned coverage rises as α falls for every model and environment, but Qwen-2.5 Game of 24 in $(2, 4]$ decreases from 79% to 72% between the 80% and 90% targets.
- Figure 1 refers to action “logits,” whereas Section 3.3 specifies frozen Sentence-BERT embeddings.
- D_{cache} denotes both the full trajectory data and the held-out calibration remainder in Section 3.3.

References

- Vishwakarma, H., Mishler, A., Cook, T., Dalmasso, N., Raman, N., & Ganesh, S. (2025). Prune ‘n Predict: Optimizing LLM decision-making with conformal prediction. *Proceedings of the 42nd International Conference on Machine Learning*, 267, 61601–61634. (proceedings.mlr.press (https://proceedings.mlr.press/v267/vishwakarma25b.html))
- Zhou, A., Yan, K., Shlapentokh-Rothman, M., Wang, H., & Wang, Y.-X. (2024). Language agent tree search unifies reasoning, acting, and planning in language models. *Proceedings of the 41st International Conference on Machine Learning*, 235, 62138–62160. (proceedings.mlr.press (https://proceedings.mlr.press/v235/zhou24r.html))

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